47202 Introduction to Future Energy - Ordinary exam - Fall 2025

Der anvendes en scoringsalgoritme, som er baseret på "One best answer"

Dette betyder følgende:
Der er altid netop ét svar som er mere rigtigt end de andre
Studerende kan kun vælge ét svar per spørgsmål
Hvert rigtigt svar giver 1 point
Hvert forkert svar giver 0 point (der benyttes IKKE negative point)

The following approach to scoring responses is implemented and is based on "One best answer"
There is always only one correct answer – a response that is more correct than the rest
Students are only able to select one answer per question
Every correct answer corresponds to 1 point
Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points)

Q1. Introduction i.

An electric motor is powered by 12 V. It spins a wheel that winds up a wire that lifts a weight of 500 g at constant speed. The electric current is 6 A. If the system works ideally without friction or electrical resistance, how fast is the weight lifted? The gravitational acceleration is 9.82 m s^{-2}

- ☐ A: 0.02 m s^{-1}
- ☐ B: 3.1 m s^{-1}
- ☐ C: 11.8 m s^{-1}
- ☐ D: 14.7 m s^{-1}
- ☐ E: 22 m s^{-1}

Solution

Velocity: $v = \frac{\mathrm{d}h}{\mathrm{d}t}$

$P = UI = 12\mathrm{V} \cdot 6\mathrm{A} = 72\mathrm{W} = 72\ \mathrm{J\ s^{-1}}$

$E_{\mathrm{pot}} = mgh$

$P = \frac{\mathrm{d}E_{\mathrm{pot}}}{\mathrm{d}t} = \frac{\mathrm{d}(mgh)}{\mathrm{d}t} = mg\frac{\mathrm{d}h}{\mathrm{d}t}$

$v = \frac{\mathrm{d}h}{\mathrm{d}t} = \frac{P}{mg} = \frac{UI}{mg} = \frac{72\ \mathrm{J\ s^{-1}}}{0.500\ \mathrm{kg} \cdot 9.82\ \mathrm{m\ s^{-2}}} =$ <u>14.7 m s^{-1}</u>

Q2. Introduction ii.

Which statement about chemical energy is most correct

Vælg en svarmulighed

- A: Chemical energy is the energy that is liberated when a chemical bond is formed or absorbed when a bond is broken
- B: Chemical energy is the energy that is needed to form a chemical bond
- C: Chemical energy is the same as internal energy
- D: Chemical energy is the energy of vibration and translation of molecules
- E: Chemical energy is energy contained in molecules. It can be calculated by Einstein's energy-mass relation, $E = mc^2$

Solution

When a chemical bond is formed, energy is liberated. Similarly, it requires energy to break a chemical bond, just like it requires energy to separate two magnets that attract each other.

Q3. Thermodynamics and electrochemistry i.
Which statement about Gibbs free energy is most correct

- ☐ A: The Gibbs free energy of a process quantifies the energy that can be utilised for mechanical or electrical work without violation of the second law of thermodynamics
- ☐ B: The Gibbs free energy of a process quantifies the total energy liberated from a process
- ☐ C: The Gibbs free energy is similar to internal energy but with an adjustment for the necessary pressure-volume work
- ☐ D: The Gibbs free energy determines reaction kinetics and thus how much work can be produced
- ☐ E: The Gibbs free energy of a process is always negative

Solution

The total energy liberated from or consumed by a chemical process at constant pressure is the enthalpy of reaction. If this is to be converted into work, either mechanical or electrical work, one needs to consider the second law of thermodynamics stating that entropy cannot be destroyed. If the process leads to decreased entropy, typically by a net reduction of the amount of gas molecules, then that amount of entropy must be generated by using part of the energy for generation of heat. The rest of the energy from the process can be used freely for anything and is therefore named free energy or Gibbs free energy.

Q4. Thermodynamics and electrochemistry ii.
Calculate the enthalpy of reaction of combustion of ethanol (C_2H_5OH)

	C_2H_5OH	O_2	CO_2	H_2O
Standard enthalpy of formation (kJ mol^{-1})	-277.6	0.0	-393.5	-237.1

- ☐ A: -353.0 kJ mol^{-1}
- ☐ B: -1220.7 kJ mol^{-1}
- ☐ C: -1994.1 kJ mol^{-1}
- ☐ D: 277.6 kJ mol^{-1}
- ☐ E: 610.4 kJ mol^{-1}

Solution

Combustion is reaction with oxygen to generate CO_2 and H_2O. In this case the reaction is

$C_2H_5OH + O_2 \rightarrow CO_2 + H_2O$ (unbalanced)

Ethanol contains 2C and 6H so we get $2CO_2$ and $3H_2O$. This requires 7O of which one is supplied by ethanol, and we thus need $3O_2$.

$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$

Enthalpy of reaction is enthalpy of formation of the product (generated) minus that of the reactants (consumed)

$$\Delta h_r^0 = 2\Delta h_{f,CO2}^0 + 3\Delta h_{f,H20}^0 - \Delta h_{f,C2H5OH}^0 - 3\Delta h_{f,O2}^0$$

$$\Delta h_r^0 = [2\cdot(-393.5) + 3\cdot(-237.1) - (-277.6) - 3\cdot 0]\ \text{kJ mol}^{-1} = \underline{1220.7\ \text{kJ mol}^{-1}}$$

Δ

Q5. Solar i.

A monocrystalline Si photovoltaic panel has an active area of 2 m^2. The panel can achieve a maximum efficiency of 20% under standard test conditions (1000 W/m^2, 25 °C, and AM 1.5). What is the maximum power output of the PV panel under standard test conditions? Neglect losses derived from other electrical components.

- ☐ A: 20 W
- ☐ B: 40 W
- ☐ C: 200 W
- ☐ D: 400 W
- ☐ E: 1000 W

Solution

From the definition of efficiency (Eq. 6 in Solar Lecture):

$$\eta = \frac{P}{P_{irradiance}} \rightarrow P = P_{irradiance} \cdot \eta$$

$$P = 1000\ \left[\frac{W}{m^2}\right] * 2[m^2] * 0.2[-] = 400\ [W]$$

Therefore, the correct answer is (D) 400 W

Q6. Solar ii.

Consider the following statements:

1. Today, the most widely installed photovoltaic cells worldwide are based on monocrystalline Si technology.
2. Global irradiance is the difference between direct and diffuse irradiances.
3. The direct solar irradiance measured at the surface of the Earth is higher than the direct solar irradiance measured at the top of the Earth's atmosphere.
4. A photon with higher frequency has higher energy than one with lower frequency.
5. The main advantage of concentrated solar power, compared to photovoltaic panels, is that it can provide electricity during the night by using thermal energy storage.

It is possible to affirm that:

- ☐ A: All statements are true
- ☐ B: 1, 3, and 4 are true
- ☐ C: 2 and 3 are false
- ☐ D: 1, 2, and 3 are false
- ☐ E: Only 3 is false

Solution

1. True. As stated in the Solar lecture slides, page 30, and tested in the Exam-like question.

2. False. Because global irradiance is the **sum** of direct and diffuse. As stated on page 13 of Solar lecture slides.

3. False. Because the direct irradiance measured at the surface of Earth is **lower** than at the top of the atmosphere. As it was discussed from statements in page 12 of Solar lecture slides

4. True. As it can be derived from the wave equations in the Solar lecture slides, page 25.

5. True. As stated in the Solar lecture slide, page 43.

Therefore, the correct answer is (C) II and III are false.

Q7. Wind i.

A three-bladed wind turbine can deliver a maximum power of 10 MW when operating under its rated wind speed of 13 m/s. If the wind turbine rotor has an area of 20000 m^2 and the density of the air is 1.2 kg/m^3, what is the power coefficient of this wind turbine operating at 13 m/s?

- ☐ A: 4%
- ☐ B: 19%
- ☐ C: 27%
- ☐ D: 38%
- ☐ E: 59%

Solution

From the turbine power curve equation, page 26 of wind Lecture:

$$P_t = C_p P_{wind} \rightarrow C_p = \frac{P_t}{P_{wind}}$$

$$C_p = \frac{P_t}{\frac{1}{2}\dot{m}v^2} = \frac{P_t}{\frac{1}{2}\rho A v^3} = \frac{10 \cdot 10^6 [W]}{\frac{1}{2} \cdot 1.2\left[\frac{kg}{m^3}\right] \cdot 20000[m^2] \cdot \left(13\left[\frac{m}{s}\right]\right)^3} \approx 0.379 \approx 38\%$$

Therefore, the correct answer is (D) 38%

Q8. Wind ii.

Consider the following statements:

1. The wind speeds for offshore regions are usually lower than for onshore regions, for the same altitudes.
2. The frequency distribution of wind speeds, over a reference year, is often represented by a Weibull distribution.
3. Wind turbines **always** produce more electricity over higher wind speeds compared to lower wind speeds.
4. Modern wind turbines control systems can adapt both the blade angle and rotor orientation for adjusting the power generation and ensuring safe operation.
5. The rated wind speed is the wind speed at which the wind turbine has maximum power generation.

It is possible to affirm:

- ☐ A: 1 and 3 are false
- ☐ B: 3 and 5 are false
- ☐ C: Only 1 is false
- ☐ D: All statements are true
- ☐ E: 1, 2, and 4 are true

Solution

1. False. The wind speeds are usually higher because the roughness of terrain is lower. As it was discussed in class from slide 12-14 in Wind lecture slides.

2. True. As stated in page 18 of Wind lecture slides.

3. False. Electricity production is limited to a constant value for wind speeds higher than the rated wind speed. As discussed in class from page 27 in Wind lecture slides.

4. True. As stated in page 37 of Wind lecture slides and tested in the Exam-like questions.

5. True. As discussed in class when explaining the wind turbine power curve, as presented in page 28 of Wind lecture slides.

Therefore, the correct alternative is A) I and III are false.

Q9. Nuclear i.
Why is a moderator needed in a usual nuclear reactor?

- A: The moderator slows down the fission process to the desired level. Without the moderator, the process would get out of control and eventually melt down
- B: The moderator is needed to slow down all the fission products to protect the reactor material and extent the lifetime
- C: The moderator damps micro vibrations in the reactor to limit wear
- D: The reactor is operated on U-235, which mostly absorbs slow neutrons (thermal neutrons). If neutrons generated by the fission process are not moderated, criticality cannot be sustained
- E: A moderator is in principle not needed at all, but is normally built in to secure smooth operation during startup and shutdown

Solution

Answer D is correct. The neutrons must be slowed down significantly to increase the probability of absorption by the uranium-235. Criticality is when the process is just maintained because at average one of the neutrons from each fission is absorbed in a new uranium nucleus.

Q10. Nuclear ii.

Calculate the mass defect of $^{239}_{9}$Pu. The atomic mass is 239.052156 amu. The mass of the subatomic particles are: neutron: 1.008665 amu, proton: 1.007276 amu and electron: 0.0005486 amu

Vælg en svarmulighed

- ☐ A: 0.02 amu
- ☐ B: 1.94 amu
- ☐ C: 3.28 amu
- ☐ D: 5.95 amu
- ☐ E: 11.7 amu

Solution

94 · 1,007276 + 145 · 1,008665 + 94 · 0,0005486 = 240,991937 amu

Real mass:	239,052156 amu
Mass defect (the difference)	1,939781 amu

Q11. Biomass i.
An anaerobic digester converts wastewater containing fructose into biogas as follows:

$C_6H_{12}O_6 \rightarrow xCH_4 + yCO_2$

Where “x” and “y” are stoichiometric coefficients that you need to determine by “balancing the reaction”. If fructose ($C_6H_{12}O_6$) is the input and methane (CH_4) is the output, what is the energy efficiency of this process on a lower heating value (LHV) basis? Assume that the LHV of fructose is 2546 kJ/mol and the LHV of methane is 802 kJ/mol. Assume complete conversion and neglect energy losses other than chemical conversion

- ☐ A: 100%
- ☐ B: 95%
- ☐ C: 63%
- ☐ D: 32%
- ☐ E: 0%

Solution

Balancing the number of carbon and oxygen atoms, one has the following reaction:

$$C_6H_{12}O_6 \rightarrow 3CH_4 + 3CO_2$$

Thus, efficiency can be calculated as:

$$\eta = \frac{3 \cdot LHV_{CH_4}}{LHV_{C_6H_{12}O_6}} = \frac{3 \cdot 802 \left[\frac{kJ}{mol}\right]}{2546 \left[\frac{kJ}{mol}\right]} \approx 0.945 \approx 95\%$$

Thus, the correct answer is B) 95%

Q12. Biomass ii.

Consider the following statements:

1. Biomass refers **only** to organic matter derived from dedicated energy crops, like wood chips, sugarcane, and maize.
2. The main advantage of biomass energy is that photosynthesis is a process more efficient than PVs in converting solar energy.
3. The pyrolysis process **always** requires oxygen for converting biomass into biochar.
4. A major criticism of bioenergy is the sustainability of using energy crops, such as wood chips, for generating electricity.
5. The gasification of biomass allows one to produce syngas ($CO+H_2$) which can be converted into chemicals.

It is possible to affirm:

- ☐ A: All statements are false
- ☐ B: 2, 4 and 5 are true
- ☐ C: Only 5 is true
- ☐ D: 4 and 5 are true
- ☐ E: Only 2 is false

Solution

1. False. Biomass also includes organic wastes, such as manure. As stated in page 5 of Biomass lecture slides.
2. False. Photosynthesis is less efficient than PVs in converting solar energy. As stated in page 12 of Biomass lecture slides.
3. False. Pyrolysis does not need oxygen. As it can be deduced from page 26 of Biomass lecture slides and discussed in class.
4. True. As stated in page 7 of Biomass lecture slides.
5. True. As stated in page 29 of Biomass lecture slides.

Thus, the correct answer is D) IV and V are true

Q13. Fuel cells and hydrogen i.
A fuel cell stack with 92 single cells is operated at 1.2 A cm^{-2} and a cell voltage of 0.78 V. The area of each electrode is 135 cm^2. What is the electrical power of the stack

- ☐ A: 126 W
- ☐ B: 8.4 kW
- ☐ C: 11.6 kW
- ☐ D: 25.5 kW
- ☐ E: 1.07 MW

Solution

92 cells · 135 cm^2 · 1.2 A cm^{-2} · 0.78 V $cell^{-1}$ = 11625 VA = 11625 W = <u>11.6 kW</u>

Q14. Fuel cells and hydrogen ii.
Which of the following issues is NOT a concern with hydrogen?

- ☐ A: Hydrogen is voluminous to store
- ☐ B: Hydrogen is poisonous
- ☐ C: Hydrogen is expensive
- ☐ D: A proper infrastructure for hydrogen is missing most places
- ☐ E: The round-trip efficiency of hydrogen (electricity-hydrogen-electricity) is inefficient

Solution

It is a convenient feature with hydrogen is that it is not poisonous. All the other points are severe concerns.

Q15. Power-to-X i.
An electrolyzer stack with 300 cells is operated at 550 V. What is the voltage efficiency (HHV). The thermoneutral voltage is 1.48 V (liquid water)

- ☐ A: 42%
- ☐ B: 55%
- ☐ C: 65%
- ☐ D: 78%
- ☐ E: 81%

Solution

300 cell at 550 V gives 1.83 V per cell.

$\eta_{\mathrm{V}} = \frac{E_{\mathrm{TN}}}{E_{\mathrm{cell}}} = \frac{1.48\ \mathrm{V}}{1.83\ \mathrm{V}} = 0.809$ = <u>81%</u>

Q16. Power-to-X ii.
What is produced by the Haber-Bosch process?

- ☐ A: Hydrogen
- ☐ B: Methanol
- ☐ C: Gasoline
- ☐ D: Diesel
- ☐ E: Ammonia

Solution

Ammonia. All ammonia today is produced by the Haber-Bosch process from gaseous hydrogen and nitrogen.

Q17. Storage i.
A flywheel with 20 kg rotating mass shaped as a massive cylinder with the radius (*R*) of 26 cm is spinning 10 000 revolutions per minute. How much energy does that represent? The angular momentum of a cylinder is: $I = ½mR^2$

- ☐ A: 0.103 kWh
- ☐ B: 1.44 kWh
- ☐ C: 5.6 kWh
- ☐ D: 10 kWh
- ☐ E: 56 kWh

Solution

$E_{\text{kin}} = ½I\omega^2$

$I = ½mR^2$

10 000 revolutions per minute = $2\pi \cdot 10\,000\ \text{min}^{-1}\,\frac{\text{min}}{60\ \text{s}} = 1047\ \text{rad s}^{-1}$

$E_{\text{kin}} = ½(½mR^2)\omega^2 = ¼mR^2\omega^2 = ¼ \cdot 20\ kg \cdot\ (0.26\ m)^2\ \cdot\ (1047\ \text{rad s}^{-1})^2 =$ 370519 J

$E_{\text{kin}} = \frac{370519\ \text{J}}{3600\ \text{Wh}\ J^{-1}} = 103$ Wh = <u>0.103 kWh</u>

Q18. Storage ii.
Which of the following groups of technologies is most suited for energy storage at a large scale

- ☐ A: Batteries and chemical storage as fuel
- ☐ B: Compressed air and flywheel
- ☐ C: Compressed air and batteries
- ☐ D: Pumped hydro and chemical storage as fuel
- ☐ E: Batteries and pumped hydro

Solution

D: Pumped hydro and chemical storage as fuel.

Batteries and flywheels are not well suited for large scale due to high cost.

Q19. Batteries i.
How does a lithium ion battery work?

- ☐ A: During discharge, metallic lithium is oxidised at the negative electrode and the ions are reduced at the positive electrode
- ☐ B: During discharge, metallic lithium is oxidised at the positive electrode and the ions are reduced at the negative electrode
- ☐ C: Positive lithium ions move back and forth and are stored in the negative electrode when the battery is charged.
- ☐ D: Negative lithium ions move back and forth and are stored in the negative electrode when the battery is charged.
- ☐ E: Lithium ions carry the current in the cell and are accumulated in the electrolyte when the battery is charged

Solution

C correct.

Re A and B: When a battery is referred to as “lithium ion”, it means that lithium is present as ions only and not metallic lithium.

Re D: Lithium ions are always positive, never negative.

Re E: lithium ions are present in the electrolyte but not accumulated, just like electrons are not accumulated in an electric wire.

Q20. Batteries ii.

A 85 kWh electric vehicle battery is fast-charged from from 20% to 80% of full charge in 20 minutes (assume constant charging current and charging voltage). What is the C-rate?

- ☐ A: 0.6 C
- ☐ B: 1.8 C
- ☐ C: 2.4 C
- ☐ D: 3.4 C
- ☐ E: 3.9 C

Solution

The rate 1 C is the rate at which the batter is fully charged or discharged in 1 hour. From 20% to 80% is 60% of the total capacity.

60% in 20 min equals 60% in (20/60) hour, which again equals 180% in an hour i.e., 1.8 C

Q21. Infrastructure i.
A fictitious high voltage AC (HVAC) transmission line operates at 110kV peak voltage and 35 kA peak current and has negligible heat losses due to electrical resistance. The grid operator measures that the real power transmitted by the line is 1500 MW. Assuming that the HVAC line is a linear circuit, how much is the phase difference between current and voltage? Assume sinusoidal steady-state operation.

- ☐ A: 15 degrees
- ☐ B: 27 degrees
- ☐ C: 39 degrees
- ☐ D: 45 degrees
- ☐ E: 59 degrees

Solution
From the definition of power factor, page 12 in Infrastructure slide, and root mean square definition, page 6, one can calculate:

$$PF = \frac{P_{real}}{V_{rms} \cdot I_{rms}} = \cos(\theta) \rightarrow \theta = \cos^{-1}\left(\frac{P_{real}}{V_{rms} \cdot I_{rms}}\right)$$

$$\theta = \cos^{-1}\left(\frac{1500\ [\mathrm{MW}]}{\frac{110}{\sqrt{2}}[\mathrm{kV}] \cdot \frac{35}{\sqrt{2}}[\mathrm{kA}]}\right) = \cos^{-1}(0.779) \approx 39\ [\mathrm{deg}]$$

Therefore, the correct answer is (C)

Q22. Infrastructure ii.

Consider the following statements:

1. The main advantage of AC electricity transmission is the ability to easily increase and/or decrease voltage by using transformers.
2. High voltage AC transmission systems often use capacitor banks to correct phase shifts caused by inductive loads.
3. Grid congestion is the inability for the electricity transmission system to handle an increase in either energy consumption or energy production.
4. The creation of a hydrogen grid could provide a more stable source of hydrogen for chemical plants.
5. Excess waste heat energy from industries can be integrated with district heating and sell heat to municipalities.

It is possible to affirm:

- ☐ A: All statements are true
- ☐ B: Only 2 is false
- ☐ C: 2 and 3 are false
- ☐ D: 2 and 4 are false
- ☐ E: Only 1 and 5 are true

Solution

1. True. As stated in page 7 of Infrastructure lecture slides and discussed in the classroom.

2. True. As stated in page 5 of the reading material, tested in the Exam-like questions (a false version using DC) and discussed in class.

3. True. As discussed from activity in page 30 of Infrastructure lecture slide.

4. True. As can be understood from page 34 in Infrastructure lecture slide and discussed in class.

5. True. As can be understood from page 38 in Infrastructure lecture slide and discussed in class.

Thus, the correct answer is (A)

Q23. Economics i.

A company was able to acquire an electrolysis system completely for free, because its efficiency for converting electricity into hydrogen energy (lower heating value basis) was only 40%, a relatively low number for the technology. If electricity can be purchased for 30 EUR/MWh, what is the levelized cost of hydrogen? Assume that the lower heating value of hydrogen is 33.33 kWh/kg and that other investment and operational costs besides electricity are negligible.

- ☐ A: 1 EUR/kg
- ☐ B: 2.5 EUR/kg
- ☐ C: 5 EUR/kg
- ☐ D: 7.5 EUR/kg
- ☐ E: 10 EUR/kg

Solution

From the levelized cost equation in page 25 of the Energy Economics slide lecture:

$$c = \frac{CAPEX \cdot f_{A/P} + OPEX_{fix}}{R_p \cdot t_{op,year}} + \sum \left(\frac{R_i}{R_p}\right) c_i - \sum \left(\frac{R_{bp}}{R_p}\right) c_{bp}$$

Since CAPEX is null, it is simplified to:

$$c_{H_2} = \frac{R_{electricity}}{R_{H_2}} c_{electricity} = \frac{LHV_{H_2}}{\eta} c_{electricity}$$

$$c_{H_2} = \frac{33.33 \left[\frac{kWh}{kg}\right]}{0.4} \left(\frac{30}{1000} \left[\frac{EUR}{kWh}\right]\right)$$

$$c_{H_2} = 2.5 \left[\frac{EUR}{kg}\right]$$

Thus, the correct answer is (B) 2.5 EUR/kg

Q24. Economics ii.

Consider the following statements:

1. Inflation only exists because of the concept of time value of money.
2. Reducing the capacity factor of a renewable energy production plant increases the costs of energy, because it increases the share of fixed costs of running the plant throughout the year.
3. The specific investment costs (i.e., $ / hydrogen production capacity) of a larger hydrogen production plant are usually lower compared to a smaller production plant, due to the economy ofscale.
4. A renewable energy plant with a lower levelized cost of energy will **Always** have a higher net present value.
5. An investment with a higher NPV will **always** be a riskier investment.

It is possible to affirm:

- ☐ A: All statements are false
- ☐ B: Only 1 is false
- ☐ C: Only 3 is true
- ☐ D: 1, 4, and 5 are true
- ☐ E: 2 and 3 are true

Solution

1. False. Inflation and time value of money are different independent concepts. As discussed in class from page 9 in Economics lecture slide and tested in the Exam-like questions (a false version of the statement).

2. True. As discussed in class from page 24 in Economics lecture slide and tested in the Exam-like questions (a false version of the statement).

3. True. As discussed from in class from page 15 of Economics lecture slide.

4. False. NPV depends on the market price and volume of sales, therefore a technology with higher costs can have a higher NPV if it sells more electricity. As discussed in class from page 28 in Economics lecture slide.

5. False. Risk and NPV are different concepts. Two investments can have the same NPV but have different risks. As discussed from pages 19 and 28 of the Economics lecture slides.

Thus, the correct answer is (E) II and III are true